

END TERM EXAMINATION
FIRST SEMESTER [B.TECH] DECEMBER 2025

Paper Code: BS-103**Subject: Applied Chemistry****Time: 3 Hours****Maximum Marks: 60**

Note: Attempt any five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Assume suitable missing data.

Q. No.	Question	Marks
Q.1	Do any five parts:	(4 × 5 = 20)
(a)	Differentiate between Higher Calorific Value (HCV) and Lower Calorific Value (LCV).	4
(b)	Calculate CaCO_3 equivalents of $\text{CaSO}_4 = 35.4 \text{ ppm}$ (MW = 136).	4
(c)	The triple point is an invariant point for a one-component system. Comment.	4
(d)	What is meant by sweetening of petrol in petroleum refining?	4
(e)	Discuss any two key technological applications of self-assembly.	4
(f)	What is Calgon conditioning in boiler feed water treatment?	4
(g)	What are biodegradable polymers? Give two examples with their chemical structures.	4
(h)	A pure metal rod half immersed in water starts corroding at the bottom. Give reason.	4
UNIT – I		
Q.2	With the help of neat diagram, explain how the calorific value of gaseous fuels can be determined by Boy's calorimeter.	5
(a)		
(b)	What are gaseous fuels? Give some examples. What are the merits and demerits of gaseous fuels over solid and liquid fuels?	3
(c)	Calculate the minimum amount of air required for complete combustion of 100 kg of fuel which has the following composition by weight: C = 80%, H = 6%, O = 5%, S = 2%, N = 2% and rest Ash = 3%.	2
OR		
Q.3	Explain knocking phenomena in petrol engine. What is cetane rating of a diesel fuel?	5
(a)		
(b)	1.26 g of a coal sample was used for nitrogen estimation by Kjeldahl's method. The evolved ammonia was collected in 20 ml of N/10 HCl. To neutralize excess acid, 6.75 ml of 0.5 N NaOH was required. Determine the percentage of nitrogen in the given coal sample.	3
(c)	What is metallurgical coke? Why is it preferred as a fuel over raw coal in blast furnaces?	2

Q. No.	Question	Marks
UNIT — II		
Q.4 (a)	Draw and discuss the phase diagram of lead-silver system. Also discuss the practical application of this system (Pattinson's process).	5
(b)	Give the monomer unit and commercial application of the following polymers: (i) Teflon (ii) Bakelite.	3
(c)	Write a short note on electroluminescent polymers and their modern optoelectronic applications.	2
OR		
Q.5 (a)	What are addition polymers? Explain using an example the cationic mechanism of chain polymer formation.	5
(b)	Discuss the effect of pressure on: (i) Transition temperature of rhombic sulfur (ii) Melting point of ice.	3
(c)	Define the following terms: (i) Component (ii) Eutectic mixture (iii) Congruent melting point.	2
UNIT — III		
Q.6 (a)	Explain the lime-soda process for removal of calcium and magnesium hardness in water.	5
(b)	A standard hard water contains 15 g of CaCO_3 per litre. 20 ml of the standard hard water solution required 25 ml of EDTA solution on titration. 100 ml of the hard water sample required 18 ml of the same EDTA solution on titration. After boiling 100 ml of this water, cooling, filtering and then titration required 12 ml of EDTA solution. Calculate the temporary and permanent hardness of water.	3
(c)	How exhausted ion-exchange resins are regenerated? Give balanced chemical reactions.	2
OR		
Q.7 (a)	What are boiler problems (scale, sludge, priming, foaming, caustic embrittlement)? How can boiler troubles be minimized?	5
(b)	Discuss the working of electro-dialysis method with the help of a labeled schematic diagram.	3
(c)	Give the ions responsible for the alkalinity in water in the following cases: (i) $P = \frac{1}{2} M$ (ii) $P > \frac{1}{2} M$ (iii) $P = M$.	2
UNIT — IV		
Q.8 (a)	Explain the mechanism of dry and wet (electrochemical) corrosion in detail.	5
(b)	What do you mean by the term "Passivity"? Explain its electrochemical significance.	3
(c)	Write an informative note on carbon nanotubes (CNTs), their structure and applications.	2
OR		
Q.9 (a)	Explain the 12 principles of Green Chemistry with suitable practical examples.	5
(b)	Discuss the various environmental and metallurgical factors influencing corrosion.	3
(c)	The nature of the oxide film decides the rate of corrosion (Pilling-Bedworth rule). Explain.	2

**** End of Question Paper ****